

SenseAI — Hardware & Infrastructure Requirements

Pre-installation checklist for school IT administrators

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Server / Dedicated Machine Requirements

Component	Minimum Requirement	Recommended	Notes
Processor (CPU)	Intel Core i7 (10th Gen+) or AMD Ryzen 7 5000+	Intel Core i9 / Xeon or AMD Ryzen 9	Minimum 8 cores, 16 threads for AI inference speed
RAM (Memory)	32 GB DDR4	64 GB DDR4/DDR5	AI model (Llama3.1:8b) alone needs ~8 GB. With Live Quiz + 40 concurrent student sessions, 32 GB provides comfortable headroom. 64 GB recommended for schools with 100+ simultaneous users
Storage (OS + Software)	512 GB SSD (NVMe preferred)	1 TB NVMe SSD	SSD mandatory — HDD causes severe AI loading delays (5–10 min per query)
Storage (AI Models)	100 GB free space	200 GB free space	Software package: ~32 GB. AI models included in package. Document storage: varies by school usage
GPU (Optional)	Not required	NVIDIA RTX 3060 / RTX 5050 (8 GB VRAM+)	GPU reduces AI response time 6–10x. Measured performance: 5 questions in ~15–44 sec (GPU, warmed up) vs 11–16 min (CPU cold)

Component	Minimum Requirement	Recommended	Notes
			start). After first job, GPU stays fast. CPU-only mode works fully. NVIDIA GPU strongly recommended for school demos
Operating System	Windows 10/11 (64-bit, Pro recommended)	Windows 11 Pro / Windows Server 2019	WSL2 must be enabled. Pro edition recommended. Home edition works if WSL2 is enabled
Network Interface	1 Gbps Ethernet (LAN)	1 Gbps Ethernet + Wi-Fi backup	Server must have a static/reserved IP on school LAN (e.g., 192.168.1.100)

✓ Tested & Recommended Machine

✓ Lenovo LOQ 2025 — Tested & Verified by SSQUARE AI SOLUTIONS

- **Processor:** AMD Ryzen 7-250 (AI Series) — 440 AI TOPS NPU
- **GPU:** NVIDIA GeForce RTX 5050 — 8 GB VRAM (CUDA 13.0) — GPU acceleration confirmed working
- **RAM:** 24 GB DDR5
- **Storage:** 1 TB NVMe SSD
- **Display:** 15.6" FHD 144Hz
- **OS:** Windows 11 Home (WSL2 enabled)
- **Model:** 83JG008NIN | **Weight:** 2.4 Kg
- **Performance (measured):** 5 questions in ~15 sec, 10 questions in ~39 sec, 11 questions in ~44 sec (after GPU warmup)
- **Cold Start Note:** First AQG job after system boot takes 11–16 minutes (GPU loading model into VRAM). All subsequent jobs are fast. Run one warmup job before any live demo or class session

⚠ Critical: Do NOT use these configurations

- Low-end or thin-and-light laptop as server — thermal throttling kills AI performance. High-performance gaming/workstation laptops (e.g., Lenovo LOQ, ASUS ROG) with dedicated GPU are acceptable
- HDD (spinning disk) storage — AI model loading takes 5–10 minutes, unusable
- Less than 32 GB RAM on CPU-only machines — AI models will crash with out-of-memory errors. Exception: machines with NVIDIA GPU (8 GB VRAM+) can run with 24 GB RAM as the model loads into GPU VRAM instead of system RAM
- Windows Home Edition — Docker Desktop requires WSL2 to be enabled; Pro edition recommended for stability
- Shared internet PC — AI inference is CPU-heavy; sharing with teacher workstation causes slowdowns

⚡ Performance Tips

Cold Start — GPU Model Warmup

After the server boots, the first AQG job takes 11–16 minutes because Ollama loads the AI model (~8 GB) into GPU VRAM for the first time. Once loaded, all subsequent jobs complete in 15–44 seconds. **Always run one dummy AQG job before a live demo or class session** to pre-warm the model. The model stays loaded as long as the system is running — no warmup needed between jobs.

Network Requirements



LAN Setup (Required)

- Switch with minimum 100 Mbps ports
- Server must have a **fixed / reserved IP**
- All teacher / student devices on same LAN
- Router must allow local traffic (no inter-VLAN blocking)
- **Live Quiz:** stable LAN essential — students answer in real-time; network drops cause missed responses
- Router/switch must not block WebSocket connections (port 80 upgrade) — all standard routers support this by default



Power & UPS

- Server must be on a **dedicated UPS**
- UPS capacity: minimum 1 kVA for server alone
- Power fluctuations cause database corruption
- Sudden shutdown during AI generation = job failure
- Recommended: APC / Luminous online UPS



Computer Lab / Client Device Requirements



Desktop / Lab PC

Any desktop or laptop with 4 GB RAM+.
No software installation needed — students open a browser and access the system directly from the school server.



LAN / Network

All lab PCs must be connected to the same local network as the server (wired LAN recommended). No internet required — SenseAI runs fully offline on the school server.



Browser Requirement

Chrome 90+ or Firefox 88+ required.
Edge Chromium supported. Internet Explorer not supported. No plugins or extensions needed.



What You Do NOT Need

- Internet connection — SenseAI is fully offline after installation
- Cloud subscription or external server — everything runs on your school server
- Client software on student/teacher PCs — just a web browser is sufficient
- IT team for daily operation — designed for non-technical school staff

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Hardware Requirements — SenseAI v1.0

Please share this with your IT administrator before purchase

SenseAI — Pre-Installation Checklist

Complete this checklist before scheduling installation

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✓ Server Readiness Checklist

Please tick each item and share this completed checklist with SSQUARE AI SOLUTIONS before your installation date.

- ☐ **Dedicated server machine ready** — not a shared workstation or laptop (i7, 32 GB RAM, 512 GB SSD minimum)
- ☐ **Windows 10/11 Pro installed** — fresh installation preferred, OS updated to latest patches
- ☐ **UPS connected to server** — minimum 1 kVA, server should survive 30-minute power cut
- ☐ **Static IP assigned to server** — e.g., 192.168.1.100 (coordinate with your network admin)
- ☐ **100 GB free disk space confirmed** — AI models, Docker images, and document storage
- ☐ **Server accessible from classroom PCs** — ping test successful from at least one teacher PC
- ☐ **Administrator account credentials ready** — needed during installation for software setup
- ☐ **Antivirus / Windows Defender exclusions set** — exclude SenseAI installation folder from real-time scanning
- ☐ **USB 3.0 port available on server** — installation is done via USB drive (minimum 64 GB, USB 3.0 required for speed)

🌐 Network Readiness Checklist

- ☐ **LAN switch with available ports** — server connected via Ethernet (not Wi-Fi)
- ☐ **All classrooms reachable on same LAN** — or at minimum teacher room + computer lab
- ☐ **Port 80 not blocked** — this is the port used by SenseAI (standard web port, same as any website)
- ☐ **Firewall allows internal traffic** — no firewall rule blocking 192.168.x.x to server IP
- ☐ **Wi-Fi available in classrooms/exam hall (optional)** — any device (laptop, tablet, smartphone) connected to the same Wi-Fi can access SenseAI through a browser — no software installation needed on the device

👤 Administrative Readiness Checklist

- ☐ **List of teachers with names and email IDs** — to create teacher accounts during setup
- ☐ **Subject structure decided** — e.g., Physics, Chemistry, Biology, Maths for Class 11
- ☐ **Student list ready (optional)** — roll number, name, class. Note: students can also join sessions using a PIN shared by the teacher — no individual student accounts required
- ☐ **Documents to upload identified** — syllabus PDFs, NCERT chapters, school policies
- ☐ **IT person designated as Tenant Admin** — one person responsible for ongoing user management

📄 Server Details to Share with SSQUARE AI SOLUTIONS

School Name _____

Server IP Address	_____ (e.g., 192.168.1.100)		
Processor	_____	RAM	_____
OS Version	_____	Storage Free	_____
TeamViewer ID	_____	TeamViewer Pass	_____
Contact Person	_____	Phone	_____
Preferred Install Date	_____		

 Ready to Install? Contact Us

Once you've completed this checklist, share it with us via email or WhatsApp. We'll schedule an installation slot (remote via TeamViewer or on-site for TN/nearby states) within 2–3 working days of receiving payment confirmation.

Email: senseai-support@ssquareai.com | **Website:** ssquareai.com

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Hardware Requirements — SenseAI v1.0

Please complete and return before installation appointment